Study Material

(Data Mining and Data Warehousing (BCA40204))

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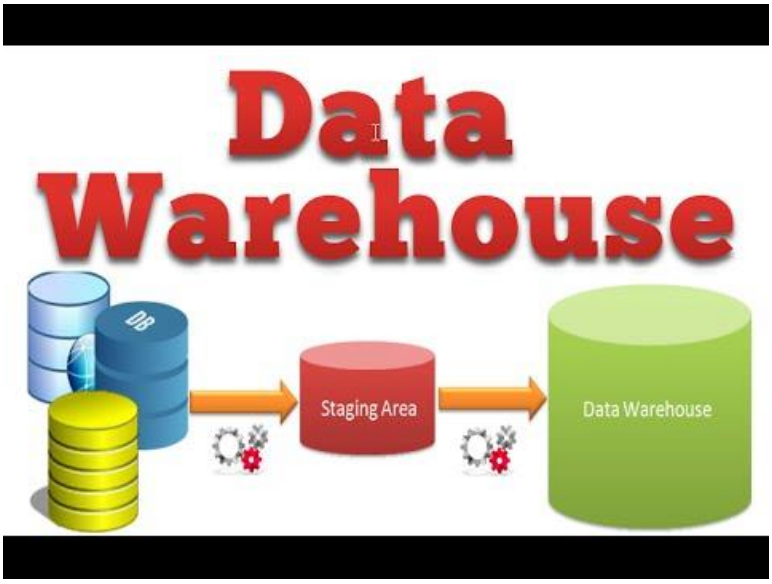
Introduction to Data WareHouse



A Data Warehouse (DW) is a relational database that is designed for query and analysis rather than transaction processing. It includes historical data derived from transaction data from single and multiple sources. A Data Warehouse provides integrated, enterprise-wide, historical data and focuses on providing support for decision-makers for data modeling and analysis.

A Data Warehouse can be viewed as a data system with the following attributes:

- It is a database designed for investigative tasks, using data from various applications.
- It supports a relatively small number of clients with relatively long interactions.
- It includes current and historical data to provide a historical perspective of information.
- Its usage is read-intensive.
- It contains a few large tables.

"Data Warehouse is a subject-oriented, integrated, and time-variant store of information in support of management's decisions."

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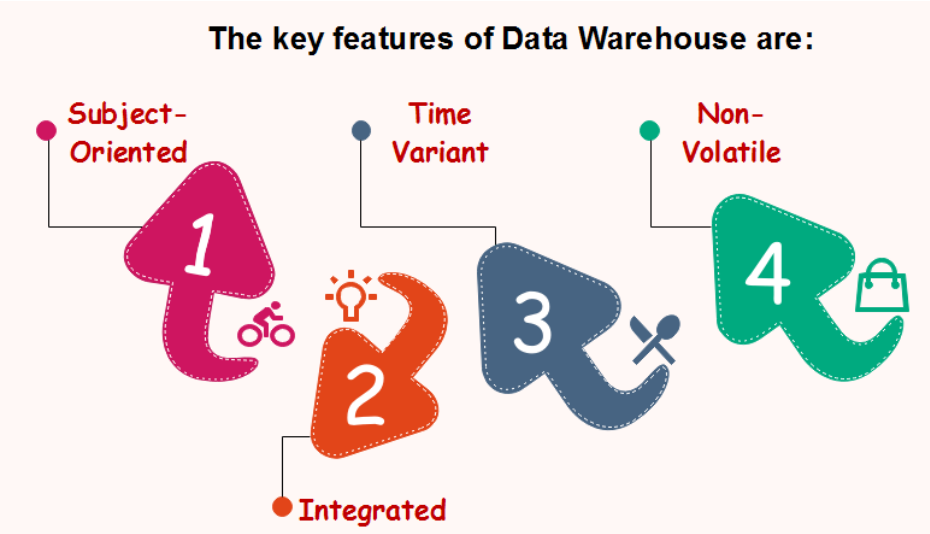
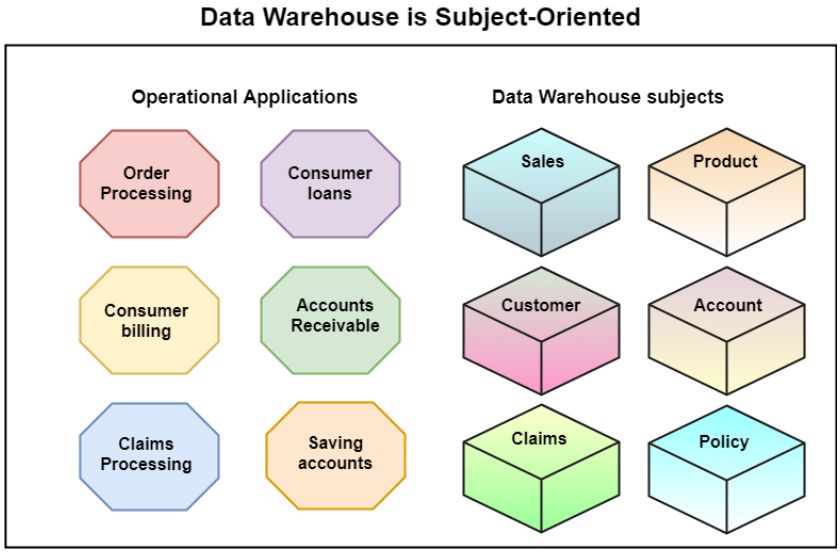


Subject-Oriented

A data warehouse target on the modeling and analysis of data for decision-makers. Therefore, data warehouses typically provide a concise and straightforward view around a particular subject, such as customer, product, or sales, instead of the global organization's ongoing operations. This is done by excluding data that are not useful concerning the subject and including all data needed by the users to understand the subject.



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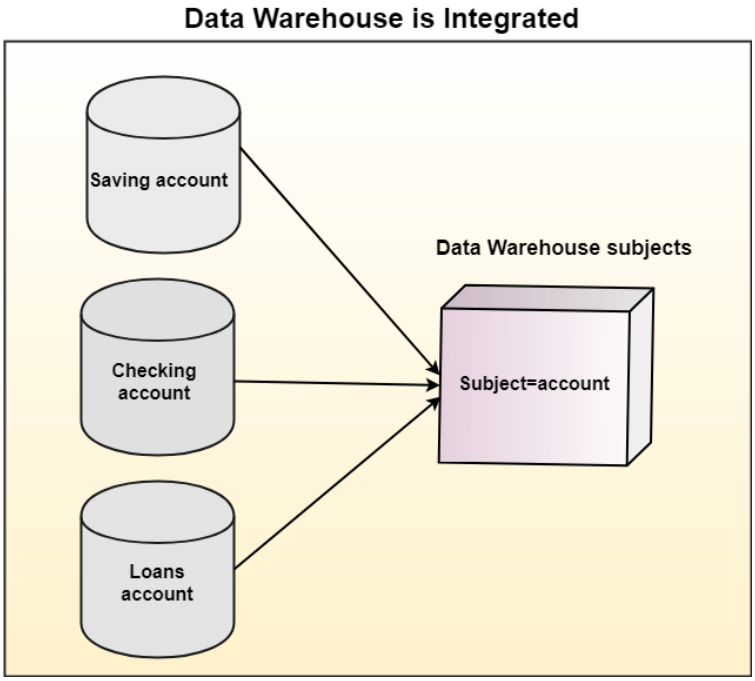
Integrated

A data warehouse integrates various heterogeneous data sources like RDBMS, flat files, and online transaction records. It requires performing data cleaning and integration during data warehousing to ensure consistency in naming conventions, attributes types, etc., among different data sources.



Time-Variant

Historical information is kept in a data warehouse. For example, one can retrieve files from 3 months, 6 months, 12 months, or even previous data from a data warehouse. These variations with a transactions system, where often only the most current file is kept.

Non-Volatile

The data warehouse is a physically separate data storage, which is transformed from the source operational RDBMS. The operational updates of data do not occur in the data warehouse, i.e., update, insert, and delete operations are not performed. It usually requires only two procedures in data accessing: Initial loading of data and access to data. Therefore, the DW does not require transaction processing, recovery, and concurrency capabilities, which allows for substantial speedup of data retrieval. Non-Volatile defines that once entered into the warehouse, and data should not change.

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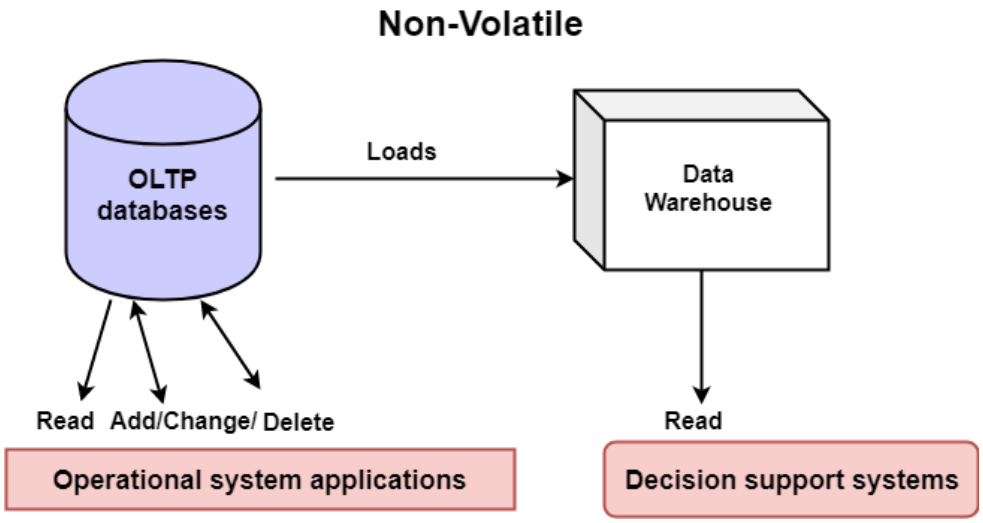

DIFFERENCES BETWEEN OPERATIONAL DATA BASE SYSTEMS AND DATA WAREHOUSE

Operational Database	Data Warehouse
Operational systems are designed to support high-volume transaction processing.	Data warehousing systems are typically designed to support high-volume analytical processing (i.e., OLAP).
Operational systems are usually concerned with current data.	Data warehousing systems are usually concerned with historical data.
Data within operational systems are mainly updated regularly according to need.	Non-volatile, new data may be added regularly. Once Added rarely changed.
It is designed for real-time business dealing and processes.	It is designed for analysis of business measures by subject area, categories, and attributes.

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It is optimized for a simple set of transactions, generally adding or retrieving a single row at a time per table.	It is optimized for extent loads and high, complex, unpredictable queries that access many rows per table.
It is optimized for validation of incoming information during transactions, uses validation data tables.	Loaded with consistent, valid information, requires no real-time validation.
It supports thousands of concurrent clients.	It supports a few concurrent clients relative to OLTP.
Operational systems are widely process-oriented.	Data warehousing systems are widely subject-oriented
Operational systems are usually optimized to perform fast inserts and updates of associatively small volumes of data.	Data warehousing systems are usually optimized to perform fast retrievals of relatively high volumes of data.
Data In	Data Out
Less Number of data accessed.	Large Number of data accessed.
Relational databases are created for on-line transactional Processing (OLTP)	Data Warehouse designed for on-line Analytical Processing (OLAP)

DATA WAREHOUSE CHARACTERISTICS

Integrated Data

One of the key characteristics of a data warehouse is that it contains integrated data. This means that the data is collected from various sources, such as transactional systems, and then cleaned, transformed, and consolidated into a single, unified view. This allows for easy access and analysis of the data, as well as the ability to track data over time.

Subject-Oriented

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A data warehouse is also subject-oriented, which means that the data is organized around specific subjects, such as customers, products, or sales. This allows for easy access to the data relevant to a specific subject, as well as the ability to track the data over time.

Non-Volatile

Another characteristic of a data warehouse is that it is non-volatile. This means that the data in the warehouse is never updated or deleted, only added to. This is important because it allows for the preservation of historical data, making it possible to track trends and patterns over time.

Time-Variant

A data warehouse is also time-variant, which means that the data is stored with a time dimension. This allows for easy access to data for specific time periods, such as last quarter or last year. This makes it possible to track trends and patterns over time.

Data Warehouse Architecture

There are three ways you can construct a data warehouse system. These approaches are classified by the number of tiers in the architecture. Therefore, you can have a:

- Single-tier architecture
- Two-tier architecture
- Three-tier architecture

Single-tier Data Warehouse Architecture

The single-tier architecture is not a frequently practiced approach. The main goal of having such an architecture is to remove redundancy by minimizing the amount of data stored.

Its primary disadvantage is that it doesn't have a component that separates analytical and transactional processing.

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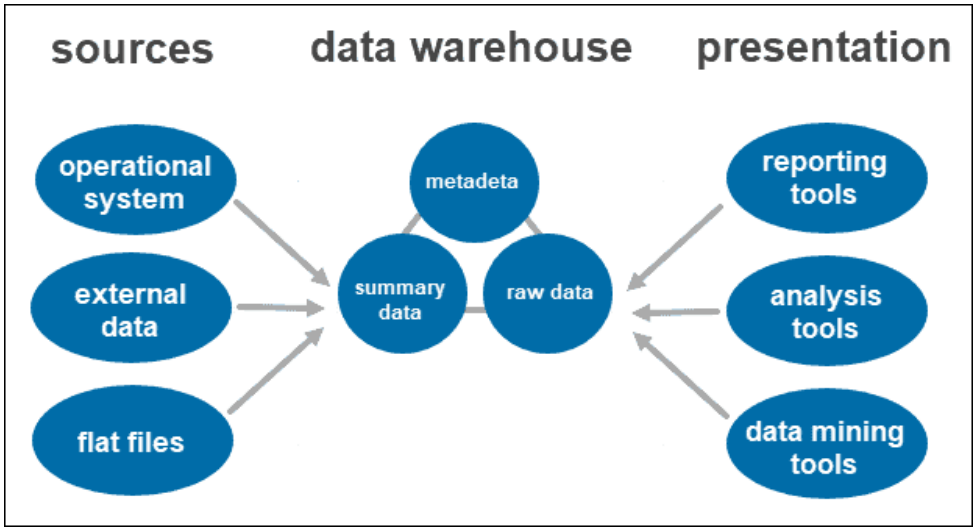
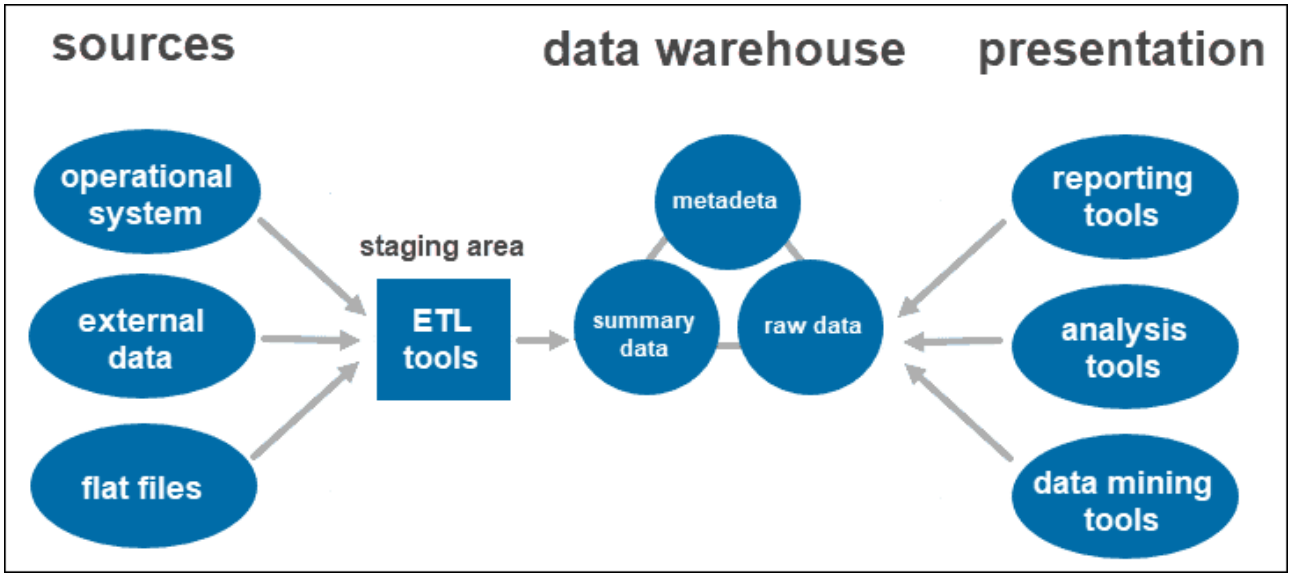


Two-tier Data Warehouse Architecture

A two-tier architecture includes a staging area for all data sources, before the data warehouse layer. By adding a staging area between the sources and the storage repository, you ensure all data loaded into the warehouse is cleansed and in the appropriate format.



This approach has certain network limitations. Additionally, you cannot expand it to support a larger number of users.

Three-tier Data Warehouse Architecture

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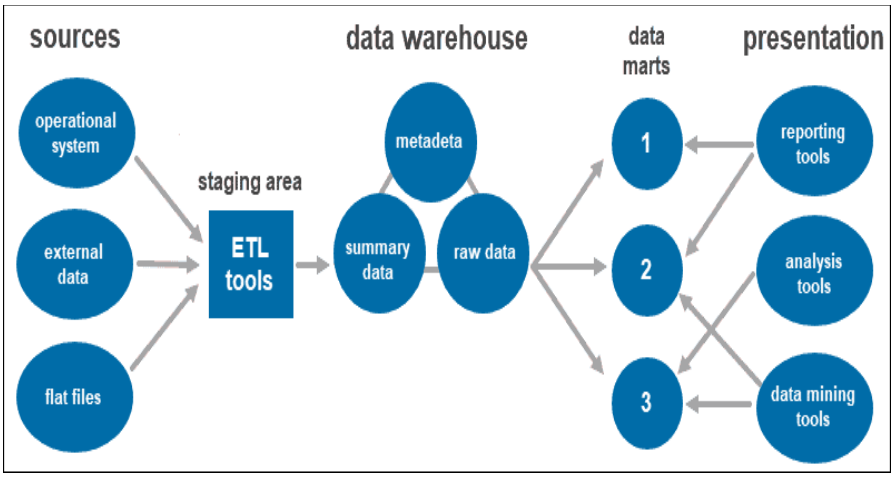
The three-tier approach is the most widely used architecture for data warehouse systems.

Essentially, it consists of three tiers:

1. **The bottom tier** is the database of the warehouse, where the cleansed and transformed data is loaded.
2. **The middle tier** is the application layer giving an abstracted view of the database. It arranges the data to make it more suitable for analysis. This is done with an OLAP server, implemented using the ROLAP or MOLAP model.
3. **The top-tier** is where the user accesses and interacts with the data. It represents the front-end client layer. You can use reporting tools, query, analysis or data mining tools.



Data Warehouse Components

From the architectures outlined above, you notice some components overlap, while others are unique to the number of tiers.

Below you will find some of the most important data warehouse components and their roles in the system.

ETL Tools

ETL stands for **Extract**, **Transform**, and **Load**. The staging layer uses **ETL tools** to extract the needed data from various formats and checks the quality before loading it into the data warehouse.

The data coming from the data source layer can come in a variety of formats. Before merging all the data collected from multiple sources into a single database, the system must clean and organize the information.

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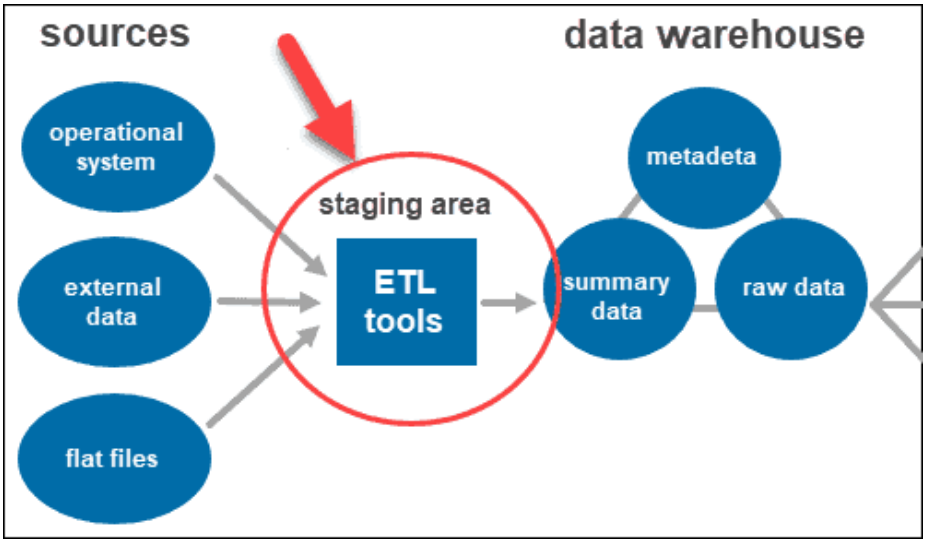

The Data warehousing Architecture Components

The most crucial component and the heart of each architecture is the database. The warehouse is where the data is stored and accessed.

When creating the data warehouse system, you first need to decide what kind of database you want to use.

There are four types of databases you can choose from:

1. **Relational databases** (row-centered databases).
2. **Analytics databases** (developed to sustain and manage analytics).
3. **Data warehouse applications** (software for data management and hardware for storing data offered by third-party dealers).
4. Cloud-based databases (hosted on the cloud).

Data

Once the system cleans and organizes the data, it stores it in the data warehouse. The data warehouse represents the central repository that stores metadata, summary data, and raw data coming from each source.

- **Metadata** is the information that defines the data. Its primary role is to simplify working with data instances. It allows data analysts to classify, locate, and direct queries to the required data.

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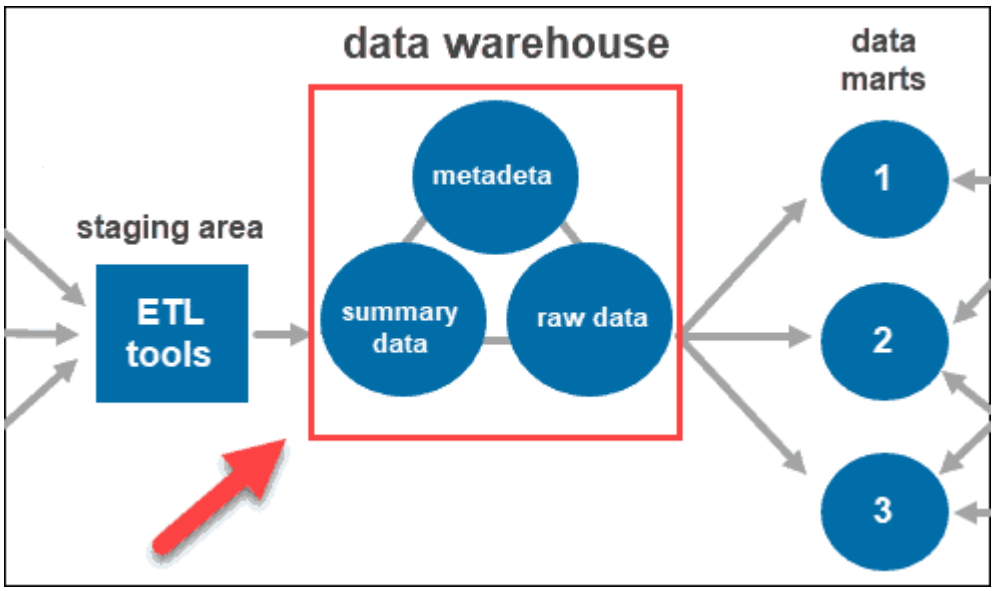
- **Summary data** is generated by the warehouse manager. It updates as new data loads into the warehouse. This component can include lightly or highly summarized data. Its main role is to speed up query performance.
- **Raw data** is the actual data loading into the repository, which has not been processed. Having the data in its raw form makes it accessible for further processing and analysis.



Access Tools

Users interact with the gathered information through different tools and technologies. They can analyze the data, gather insight, and create reports.

Some of the tools used include:

- **Reporting tools.** They play a crucial role in understanding how your business is doing and what should be done next. Reporting tools include visualizations such as graphs and charts showing how data changes over time.
- **OLAP tools.** Online analytical processing tools which allow users to analyze multidimensional data from multiple perspectives. These tools provide fast processing and valuable analysis. They extract data from numerous relational data sets and reorganize it into a multidimensional format.
- **Data mining tools.** Examine data sets to find patterns within the warehouse and the correlation between them. Data mining also helps establish relationships when analyzing multidimensional data.

Data Marts

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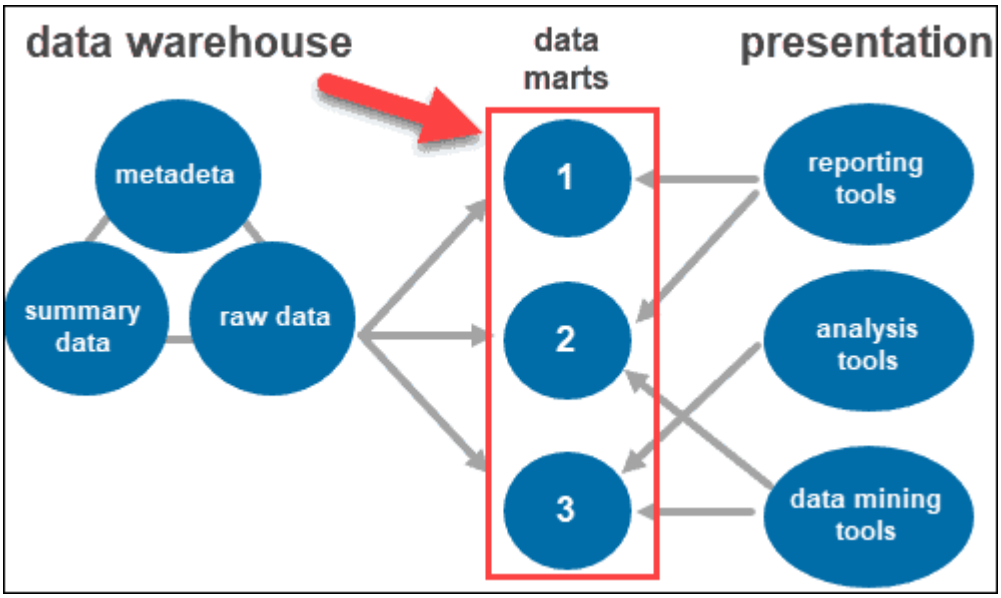
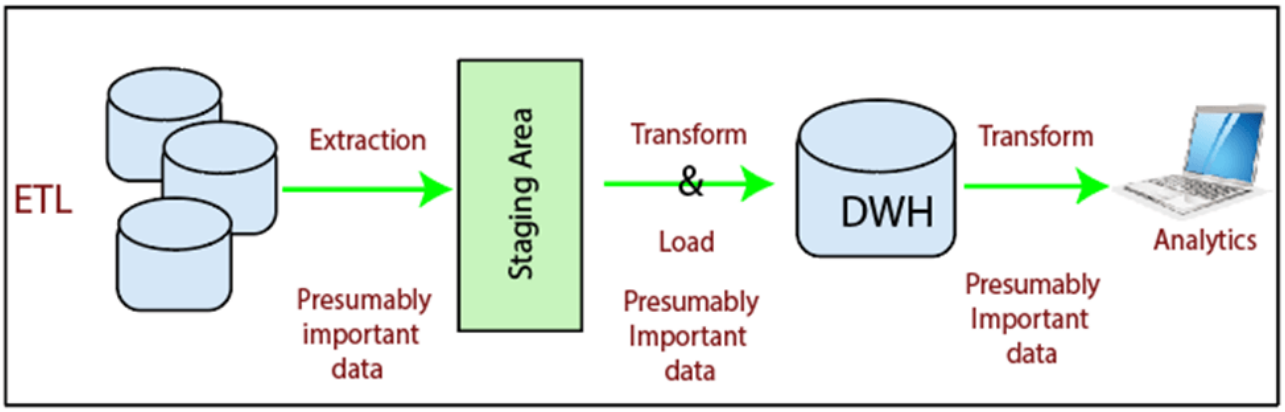
Data marts allow you to have multiple groups within the system by segmenting the data in the warehouse into categories. It partitions data, producing it for a particular user group.

For instance, you can use data marts to categorize information by departments within the company.



Extraction Transformation-Loading



Extraction

- Extraction is the operation of extracting information from a source system for further use in a data warehouse environment. This is the first stage of the ETL process.

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- Extraction process is often one of the most time-consuming tasks in the ETL.
- The source systems might be complicated and poorly documented, and thus determining which data needs to be extracted can be difficult.
- The data has to be extracted several times in a periodic manner to supply all changed data to the warehouse and keep it up-to-date.

Cleansing

The cleansing stage is crucial in a data warehouse technique because it is supposed to improve data quality. The primary data cleansing features found in ETL tools are rectification and homogenization. They use specific dictionaries to rectify typing mistakes and to recognize synonyms, as well as rule-based cleansing to enforce domain-specific rules and defines appropriate associations between values.

The following examples show the essential of data cleaning:

If an enterprise wishes to contact its users or its suppliers, a complete, accurate and up-to-date list of contact addresses, email addresses and telephone numbers must be available.

If a client or supplier calls, the staff responding should be quickly able to find the person in the enterprise database, but this need that the caller's name or his/her company name is listed in the database.

If a user appears in the databases with two or more slightly different names or different account numbers, it becomes difficult to update the customer's information.

Transformation

Transformation is the core of the reconciliation phase. It converts records from its operational source format into a particular data warehouse format. If we implement a three-layer architecture, this phase outputs our reconciled data layer.

The following points must be rectified in this phase:

- Loose texts may hide valuable information. For example, XYZ PVT Ltd does not explicitly show that this is a Limited Partnership company.
- Different formats can be used for individual data. For example, data can be saved as a string or as three integers.

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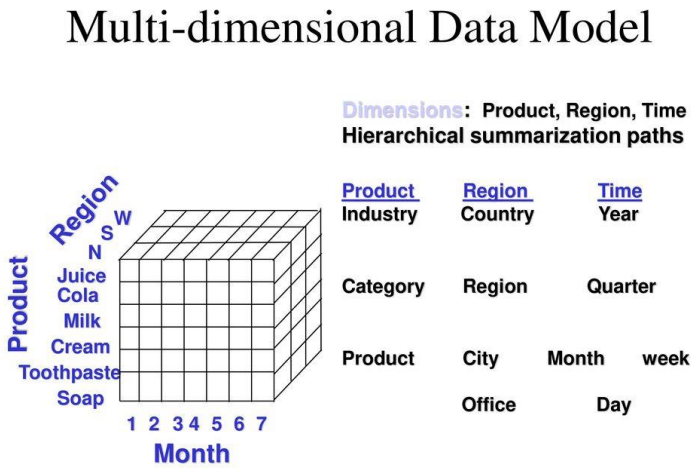
LOGICAL (MULTIDIMENSIONAL)

In the logical multidimensional model, a cube represents all measures with the same shape, that is, the exact same dimensions. In a cube shape, each edge represents a dimension. The dimension members are aligned on the edges and divide the cube shape into cells in which data values are stored.



"Logical (Multidimensional)" refers to the logical organization and representation of data in a multidimensional model. This model is used to structure data in a way that is optimized for analytical processing and reporting, making it easier for users to perform complex queries and analysis.

Here are some key concepts related to Logical (Multidimensional) data modeling in data warehousing:

1. Multidimensional Model: A multidimensional model organizes data into multiple dimensions, where each dimension represents a specific aspect or attribute of the data. For example, in a sales data warehouse, dimensions might include Time, Product, Geography, and Customer.
2. Dimensions: Dimensions are the categories or characteristics by which data can be analyzed. Each dimension consists of a set of hierarchies, attributes, and levels. In the sales data example, the Time dimension might have hierarchies like Year > Quarter > Month > Day.
3. Hierarchies: Hierarchies represent the relationship between various levels within a dimension. They enable drill-down and roll-up operations for data analysis. For instance, within the Time dimension, you can drill down from Year to Quarter, Month, and Day.
4. Measures: Measures are the numeric data points that represent the key performance indicators (KPIs) in the data warehouse. These are the values that users want to analyze, such as sales revenue, quantity sold, or profit.

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5. Cubes: A logical multidimensional data model is typically organized as a cube. A cube is a collection of measures and the dimensions associated with those measures. It allows users to perform slicing, dicing, drilling, and pivoting operations for analysis.
6. Slicing and Dicing: Slicing refers to viewing a subset of data by fixing one or more dimensions. Dicing, on the other hand, involves viewing a subset of data by selecting specific values from multiple dimensions. These operations help in narrowing down data to analyze specific aspects of interest.
7. Drilling and Pivoting: Drilling involves navigating through the hierarchy levels of a dimension to get more detailed or summarized data. Pivoting involves rotating the cube to view data from different perspectives.

DATA MODELING

Data modeling in data warehousing is a crucial process that involves designing the structure and organization of data to support effective storage, retrieval, and analysis within a data warehouse. Data modeling in this context helps to create a blueprint of the data warehouse, making it easier to understand and maintain.

Here are the key aspects and steps involved in data modeling for data warehousing:

1. Requirements Gathering: The first step in data modeling is to gather requirements from various stakeholders, including business users, analysts, and subject matter experts. Understanding their data needs and reporting requirements is essential to design a data warehouse that meets their expectations.
2. Conceptual Data Model: The conceptual data model represents high-level business entities, their attributes, and the relationships between them. It provides a clear understanding of the business domain and acts as a foundation for the subsequent modeling steps.
3. Logical Data Model: The logical data model translates the conceptual model into a technology-agnostic representation of the data. It defines the entities, attributes, and relationships in detail and helps in defining the primary keys, foreign keys, and data types.
4. Dimensional Data Model: In data warehousing, the dimensional data model is a common approach used to organize data efficiently for analytical purposes. It involves creating dimensional models with facts (numeric measures) and dimensions (attributes by which data is analyzed) to build a star schema or snowflake schema.
   - Star Schema: In a star schema, the central fact table is connected to multiple dimension tables, each representing a specific category of data. The fact table contains the measures, and dimension tables contain attributes that describe the data in the fact table.
   - Snowflake Schema: The snowflake schema is an extension of the star schema where dimension tables are normalized into multiple related tables. This reduces data redundancy but may increase query complexity.
5. Physical Data Model: The physical data model maps the logical data model to the specific database technology used for the data warehouse. It defines the physical storage structures, indexes, partitions, and optimization techniques to enhance query performance.
6. Data Integration: Data modeling in data warehousing also involves addressing the process of integrating data from various source systems into the data warehouse. This step includes data extraction, transformation, and loading (ETL) processes to ensure that data is cleansed, transformed, and loaded accurately.

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7. Metadata Management: Metadata, which provides information about the data in the data warehouse, plays a vital role in data modeling. Proper metadata management helps users understand the data, its source, and its meaning.
8. Maintenance and Evolution: Data modeling is not a one-time task. As the business evolves and data requirements change, the data model must be updated and maintained accordingly to ensure the data warehouse remains aligned with business needs.

Overall, data modeling in data warehousing is a comprehensive process that involves understanding business requirements, designing efficient data structures, and ensuring data quality to create a powerful foundation for data analysis and decision-making.

SCHEMA DESIGN

Schema design is the process of structuring and organizing the database to efficiently store, retrieve, and manage data in a way that meets the requirements of the application or system. It is a critical step in database development and is essential for maintaining data integrity, optimizing performance, and facilitating data analysis and reporting. The schema design process involves several key considerations:

1. Requirements Gathering: Understand the data requirements of the application or system. This involves gathering information from stakeholders, including business users, analysts, and subject matter experts, to identify the entities, attributes, relationships, and constraints that the database must support.
2. Entity-Relationship Modeling: Use an entity-relationship (ER) diagram to represent the conceptual model of the database. In an ER diagram, entities represent real-world objects or concepts, and relationships define the associations between entities.
3. Normalization: Apply normalization techniques to ensure data integrity and minimize redundancy. Normalization involves breaking down data into smaller, well-organized tables to reduce data duplication and update anomalies. The result is a set of normalized tables that conform to specific normal forms (e.g., first normal form, second normal form, etc.).
4. Denormalization (Optional): In some cases, denormalization may be applied to improve performance in read-heavy applications. Denormalization involves intentionally introducing redundancy into the schema to reduce the need for joins and improve query performance.
5. Data Types and Constraints: Choose appropriate data types for each attribute to ensure data accuracy and efficient storage. Apply constraints such as primary keys, foreign keys, unique constraints, and check constraints to maintain data integrity.

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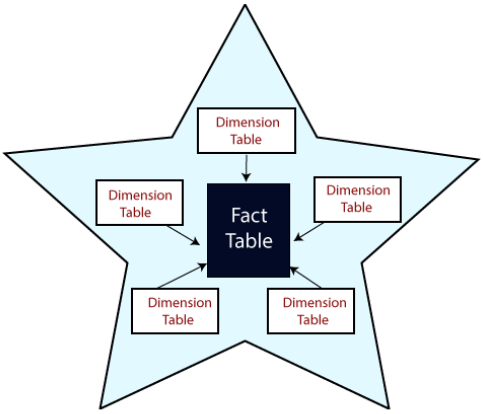
6. Indexing: Determine which columns will be indexed to speed up data retrieval. Indexes improve query performance by allowing the database to locate data more quickly.
7. Data Integrity: Enforce data integrity through referential integrity constraints and other business rules. This ensures that data remains consistent and valid across related tables.
8. Performance Optimization: Optimize the schema design for performance. Consider the types and frequency of queries that will be executed and design the schema to support those queries efficiently.
9. Scalability: Plan for future growth and scalability by designing the schema to accommodate increasing amounts of data and users.
10. Security: Implement appropriate access controls and security measures to protect sensitive data.
11. Documentation: Document the schema design thoroughly, including entity definitions, relationships, attributes, constraints, and any other relevant information.
12. Testing: Thoroughly test the schema design to ensure it meets the requirements and performs as expected.

Schema design is a crucial step in the database development lifecycle, and a well-designed schema forms the foundation for an efficient and reliable database system.

STAR AND SNOW-FLAKE SCHEMA:

The Star Schema is a widely used schema design in data warehousing due to its simplicity and ease of use for analytical queries. In the Star Schema, the database is organized into a central fact table connected to multiple dimension tables. The fact table contains the measures (quantitative data) that users want to analyze, and the dimension tables contain descriptive attributes by which the data can be analyzed.



Star Schema

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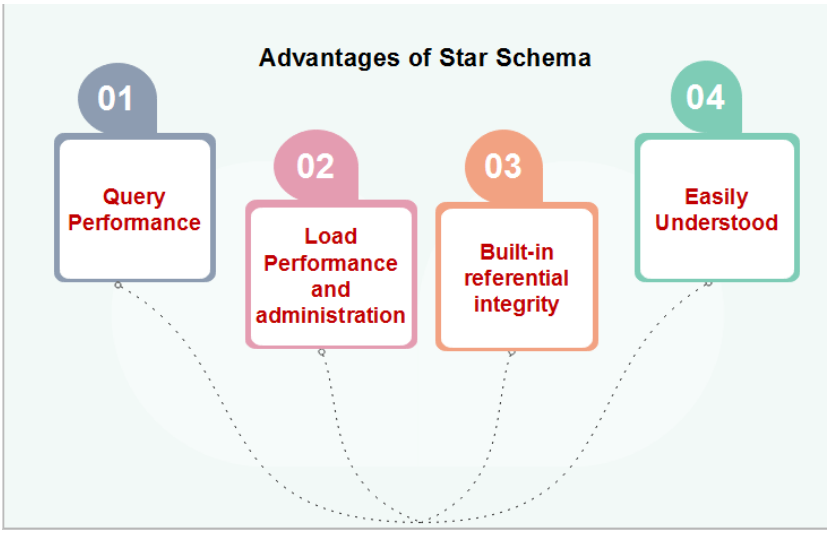
Key characteristics of the Star Schema:

- Fact Table: The central fact table contains the quantitative measures or metrics. For example, in a sales data warehouse, the fact table may contain measures such as sales revenue, quantity sold, and profit.
- Dimension Tables: Each dimension table represents a specific category of data, such as Time, Product, Customer, or Geography. The dimension tables contain attributes that describe the data in the fact table. For instance, the Time dimension may have attributes like Year, Quarter, Month, and Day.
- Denormalization: The dimension tables are usually denormalized, meaning that redundant data is intentionally introduced to improve query performance. This redundancy simplifies queries and reduces the need for complex joins, resulting in faster analytical processing.
- Query Performance: The Star Schema's denormalized design leads to better query performance, as it requires fewer joins between tables compared to a fully normalized schema like the Snowflake Schema.

ADVANTAGES OF STAR SCHEMA

Star Schemas are easy for end-users and application to understand and navigate. With a well-designed schema, the customer can instantly analyze large, multidimensional data sets.

The main advantage of star schemas in a decision-support environment are:



SNOW-FLAKE SCHEMA

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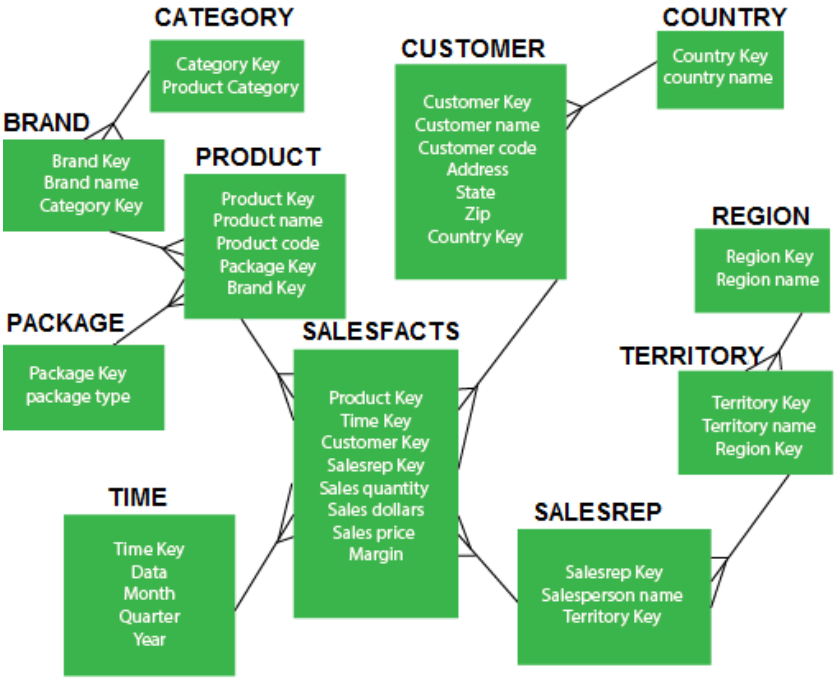
The Snowflake Schema is an extension of the Star Schema that addresses the issue of data redundancy by normalizing the dimension tables. In this design, the dimension tables are broken down into multiple related tables to eliminate redundancy. This approach is particularly beneficial when dealing with large dimensions with many attributes.



Snowflake Schema

A snowflake schema is designed for flexible querying across more complex dimensions and relationship. It is suitable for many to many and one to many relationships between dimension levels.

Advantage of Snowflake Schema

1. The primary advantage of the snowflake schema is the development in query performance due to minimized disk storage requirements and joining smaller lookup tables.
2. It provides greater scalability in the interrelationship between dimension levels and components.
3. No redundancy, so it is easier to maintain.

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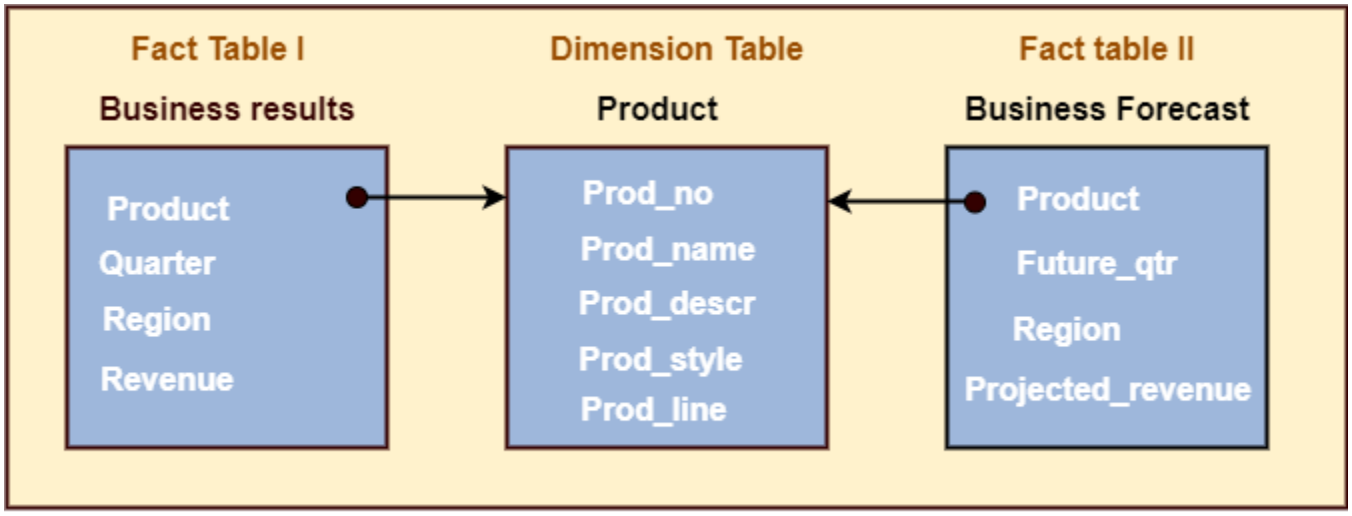
Disadvantage of Snowflake Schema

1. The primary disadvantage of the snowflake schema is the additional maintenance efforts required due to the increasing number of lookup tables. It is also known as a multi fact star schema.
2. There are more complex queries and hence, difficult to understand.
3. More tables more join so more query execution time.

FACT CONSTELLATION:

A Fact constellation means two or more fact tables sharing one or more dimensions. It is also called **Galaxy schema**.

Fact Constellation Schema describes a logical structure of data warehouse or data mart. Fact Constellation Schema can design with a collection of de-normalized FACT, Shared, and Conformed Dimension tables.



FACT Constellation Schema

Fact Constellation Schema is a sophisticated database design that is difficult to summarize information. Fact Constellation Schema can implement between aggregate Fact tables or decompose a complex Fact table into independent simplex Fact tables.

Example: A fact constellation schema is shown in the figure below.

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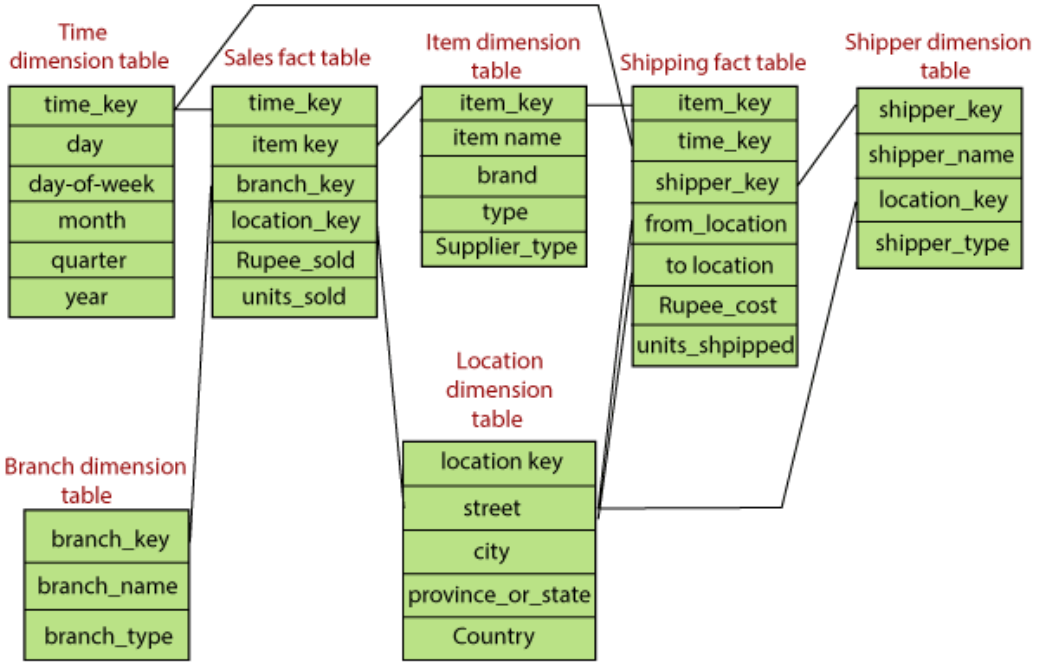

This schema defines two fact tables, sales, and shipping. Sales are treated along four dimensions, namely, time, item, branch, and location. The schema contains a fact table for sales that includes keys to each of the four dimensions, along with two measures: Rupee_sold and units_sold. The shipping table has five dimensions, or keys: item_key, time_key, shipper_key, from_location, and to_location, and two measures: Rupee_cost and units_shipped.

The primary disadvantage of the fact constellation schema is that it is a more challenging design because many variants for specific kinds of aggregation must be considered and selected.

FACT TABLE:

A Fact Table is a central table in a star schema of a data warehouse. It is an important concept required for ***Data Warehousing and BI Certification***. A fact table stores quantitative information for analysis and is often denormalized. A fact table works with dimension tables and it holds the data to be analyzed and a dimension table stores data about the ways in which the data can be analyzed.

Thus, a fact table consists of two types of columns. The foreign keys column allows to join with dimension tables and the measure columns contain the data that is being analyzed.

Types of Fact Table

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1. Transaction Fact Table

The transaction fact table is a basic approach to operate the businesses. These fact tables represent an event that occurs at the primary point. A line exists in the fact table for the customer or product when the transaction occurs.

Many rows in a fact table connect to a customer or product because they are involved in multiple transactions. Transaction data is often structured quickly in a one-dimensional framework. The lowest-level data is the rawest dimensional data that cannot be done by summarized data.

2. Snapshot Fact Table

The snapshot fact table describes the state of things at a particular time and contains many **semi-additive** and **non-additive** facts.

Example: The daily equilibrium fact is expressed by the customer dimension but not by the time dimension.

Periodic snapshots require the performance of the business at regular and estimated time intervals. Unlike a transaction fact table where we load a row for each event, with periodic snapshots, we take a picture of the activity at the end of the day, week, or month, and then another picture at the end of the next period.

Example: Performance summary of a salesman during the previous month.

3. Accumulated Fact Sheet

The accumulated fact table is used to show the activity of a process that has a beginning and an end.

For example, we are processing an order. An order remains in the process until it will be processed. As the step towards completing the order is completed, the corresponding row in the fact table is updated.

FULLY ADDITIVE, SEMI-ADDITIVE, AND NON-ADDITIVE MEASURES:

1. Fully Additive Measures: Fully Additive measures are those that can be aggregated across all dimensions without any loss of information or change in their meaning. When aggregated, these measures retain their properties, and the results make sense at any level of granularity. Examples of fully additive measures include:

- Sum of Sales Revenue: Regardless of the dimension used (e.g., Time, Product, Geography), the sum of sales revenue for a specific time period, product category, or geographic region is meaningful and accurate.
- Quantity Sold: The total quantity sold is fully additive since it retains its meaning when aggregated across various dimensions.
- Profit: The sum of profits for different products, time periods, or regions is meaningful and consistent.

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2. Semi-Additive Measures: Semi-Additive measures are those that can be aggregated across some dimensions but not all. When aggregated along certain dimensions, these measures make sense, but for other dimensions, aggregation may not be appropriate. Examples of semi-additive measures include:

- Account Balance: The account balance is semi-additive because it is meaningful to sum the balances for specific accounts, but it is not meaningful to sum the balances across time. Adding account balances across time would not represent the true financial position.
- Stock Quantity: The quantity of stock on hand is semi-additive, as it is meaningful to sum the stock quantities for specific products but not for different points in time.

3. Non-Additive Measures: Non-Additive measures are those that do not make sense when aggregated across any dimension. These measures do not retain their properties when summed or aggregated. Examples of non-additive measures include:

- Ratios and Percentages: Ratios and percentages, such as profit margins or market share, are non-additive measures. Summing these measures across dimensions would not yield meaningful results.
- Average: The average of a measure is non-additive. Aggregating averages across dimensions would not provide an accurate overall average.

Understanding the additive properties of measures is crucial in data modeling to ensure that Fact Tables are designed appropriately for accurate and meaningful data analysis. When building data cubes or performing OLAP operations, the nature of the measures guides how they can be aggregated and used to gain insights into the data.

"FACT LESSFACTS" AND "DIMENSION TABLE CHARACTERISTICS":

1. Facts: In data warehousing, Facts represent the quantitative and numerical data that businesses want to analyze. They are the measurable data points and key performance indicators (KPIs) related to the business processes or events being tracked. Examples of facts include sales revenue, quantity sold, profit, expenses, and customer counts. Facts are typically stored in the Fact Table within a data warehouse and are associated with dimension tables through foreign keys. Facts are often additive or semi-additive, allowing them to be aggregated along various dimensions to perform data analysis.
2. Dimension Tables: Dimension Tables contain descriptive attributes by which the facts are analyzed. They provide context to the facts and enable users to slice and dice data to gain insights from different perspectives. Dimension tables represent the "who," "what," "where," "when," and "how" aspects of the data. For example, in a sales data warehouse, dimension tables might include Time (representing various time periods like date, month, quarter, year), Product (representing various products with their attributes), Customer (representing various customers with their attributes), and Geography (representing various geographical regions or locations).

Characteristics of Dimension Tables:

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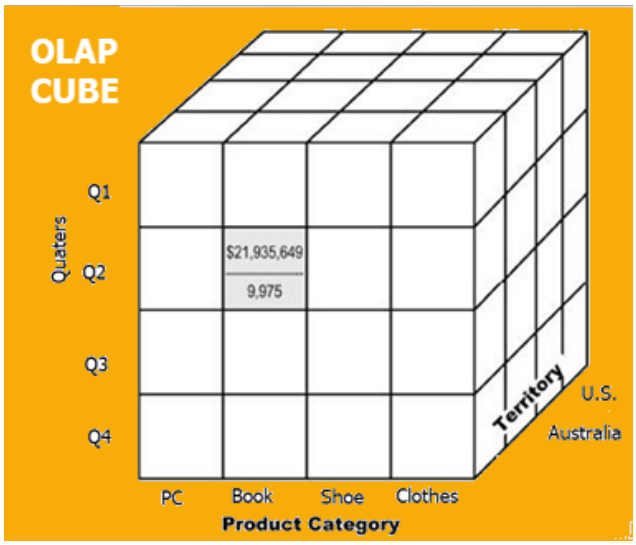
- Unique Key: Each row in a dimension table should have a unique key, which is used to establish relationships with the fact table.
- Denormalization: Dimension tables are often denormalized to reduce the need for complex joins during queries, improving performance.
- Descriptive Attributes: Dimension tables contain descriptive attributes that provide additional context and details about the facts. These attributes are used for slicing and dicing data during analysis.
- Hierarchies: Dimension tables can have hierarchies that represent the levels of detail within the dimension. For example, the Time dimension might have a hierarchy like Year > Quarter > Month > Day.
- Slowly Changing Dimensions (SCDs): Dimension tables might include historical data and require management of changes over time, using techniques like Slowly Changing Dimensions (SCDs) to maintain historical records.
- Surrogate Keys: In some cases, dimension tables use surrogate keys (auto-generated keys) in addition to natural keys to efficiently handle updates and maintain data integrity.

OLAP CUBE:

Online Analytical Processing (OLAP) is a category of software that allows users to analyze information from multiple database systems at the same time.



OLAP Cube
At the core of the OLAP concept, is an OLAP Cube. The OLAP cube is a data structure optimized for very quick data analysis.

The OLAP Cube consists of numeric facts called measures which are categorized by dimensions. OLAP Cube is also called the **hypercube**.

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Usually, data operations and analysis are performed using the simple spreadsheet, where data values are arranged in row and column format. This is ideal for two-dimensional data. However, OLAP contains multidimensional data, with data usually obtained from a different and unrelated source. Using a spreadsheet is not an optimal option. The cube can store and analyze multidimensional data in a logical and orderly manner.

How does it work?

A Data warehouse would extract information from multiple data sources and formats like text files, excel sheet, multimedia files, etc.

The extracted data is cleaned and transformed. Data is loaded into an OLAP server (or OLAP cube) where information is pre-calculated in advance for further analysis.

Basic analytical operations of OLAP

Four types of analytical OLAP operations are:

1. Roll-up
2. Drill-down
3. Slice and dice
4. Pivot (rotate)

1) Roll-up:

Roll-up is also known as “consolidation” or “aggregation.” The Roll-up operation can be performed in 2 ways

1. Reducing dimensions
2. Climbing up concept hierarchy. Concept hierarchy is a system of grouping things based on their order or level.

Consider the following diagram

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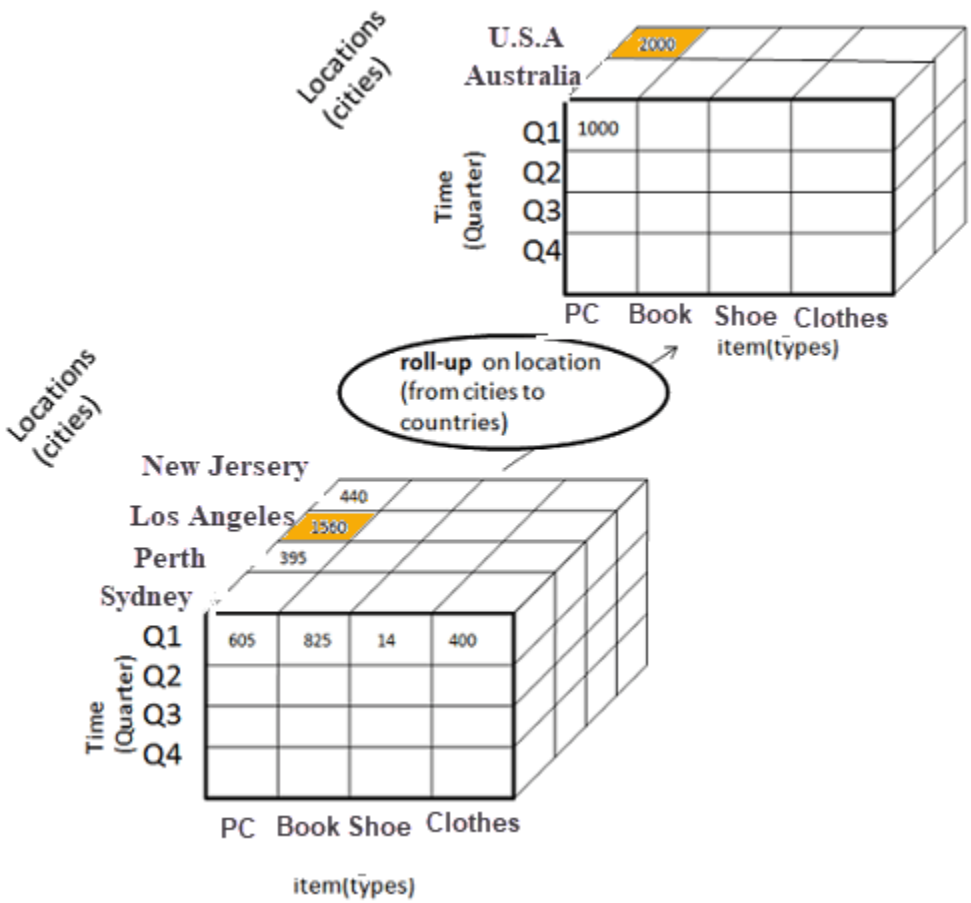

Roll-up operation in OLAP

- In this example, cities New jersey and Lost Angles and rolled up into country USA
- The sales figure of New Jersey and Los Angeles are 440 and 1560 respectively. They become 2000 after roll-up
- In this aggregation process, data is location hierarchy moves up from city to the country.
- In the roll-up process at least one or more dimensions need to be removed. In this example, Cities dimension is removed.

2) Drill-down

In drill-down data is fragmented into smaller parts. It is the opposite of the rollup process. It can be done via

- Moving down the concept hierarchy
- Increasing a dimension

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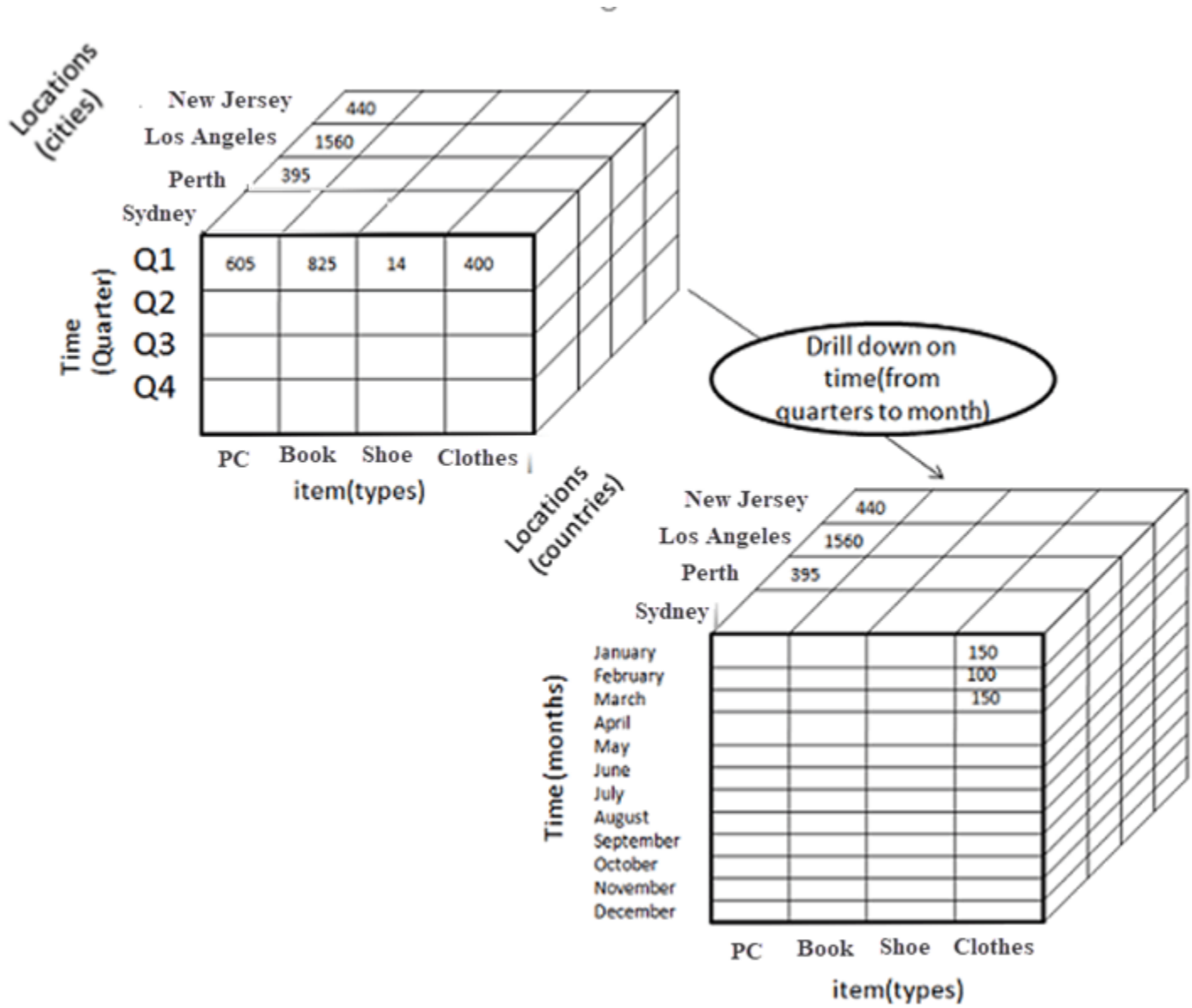

Drill-down operation in OLAP
Consider the diagram above

- Quater Q1 is drilled down to months January, February, and March. Corresponding sales are also registers.
- In this example, dimension months are added.

3) Slice:

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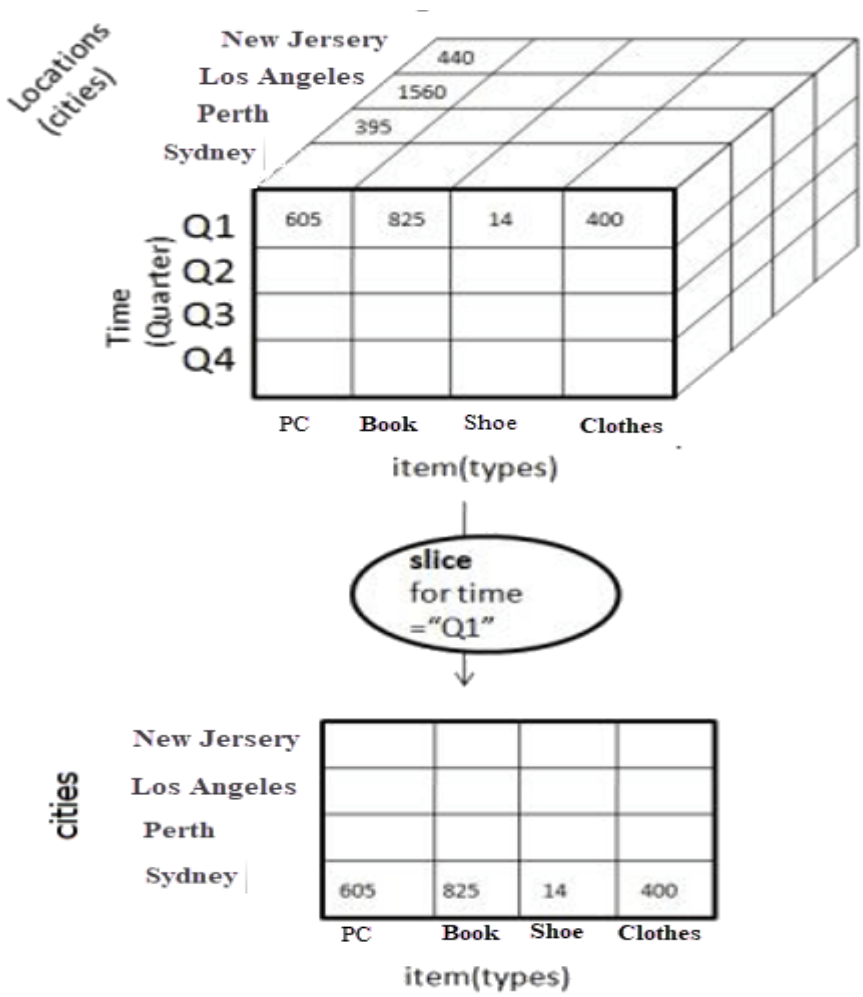
Here, one dimension is selected, and a new sub-cube is created.

Following diagram explain how slice operation performed:



Slice operation in OLAP

- Dimension Time is Sliced with Q1 as the filter.
- A new cube is created altogether.

Dice:

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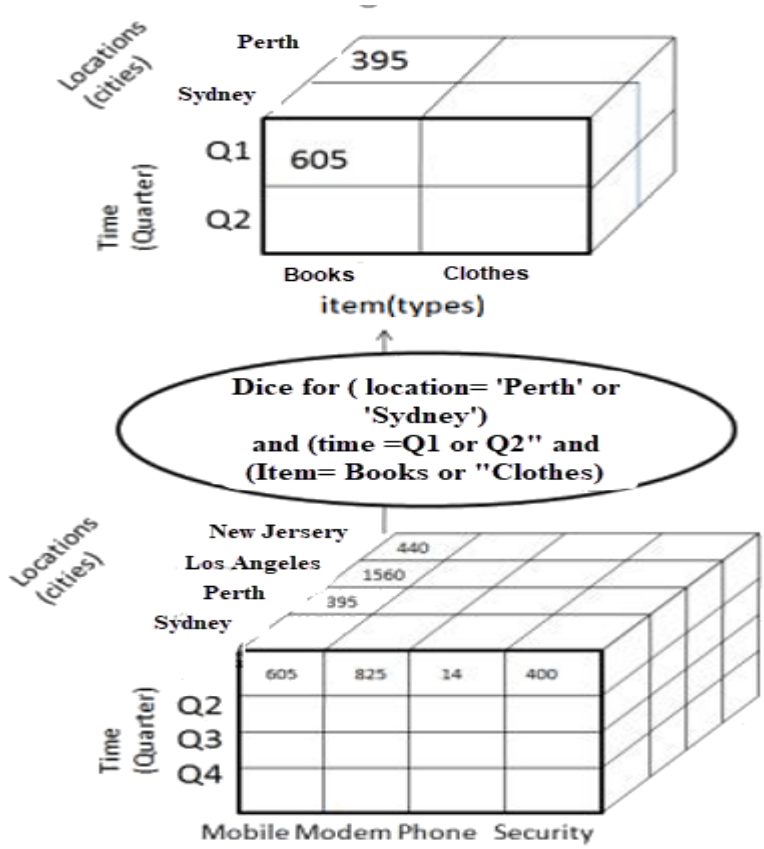
This operation is similar to a slice. The difference in dice is you select 2 or more dimensions that result in the creation of a sub-cube.



Dice operation in OLAP

4) Pivot

In Pivot, you rotate the data axes to provide a substitute presentation of data.

In the following example, the pivot is based on item types.

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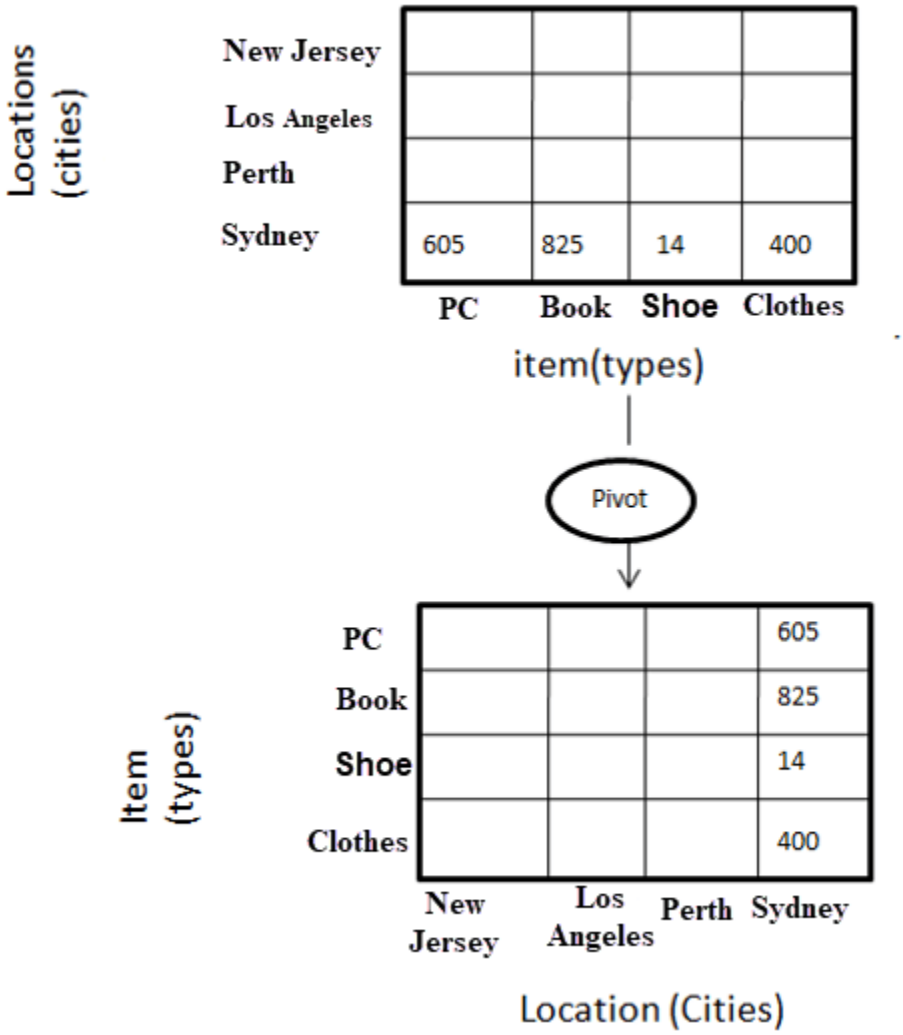

Pivot operation in OLAP

Types of OLAP systems

OLAP Hierarchical Structure

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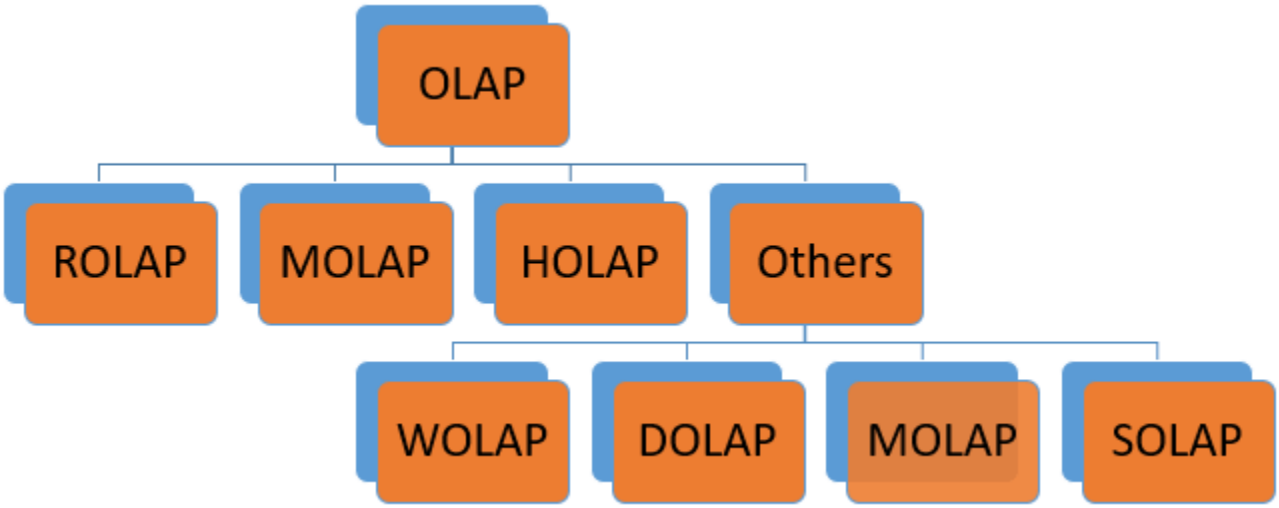

Types of OLAP Systems

Type of OLAP	Explanation
Relational OLAP(ROLAP):	ROLAP is an extended RDBMS along with multidimensional data mapping to perform the relational operation.
Multidimensional OLAP (MOLAP)	MOLAP Implementes operation in multidimensional data.
Hybrid OnlineAnalytical Processing (HOLAP)	In HOLAP approach the aggregated totals are stored in a multidimensional database w detailed data is stored in the relational database. This offers both data efficiency of the ROLA and the performance of the MOLAP model.
Desktop OLAP (DOLAP)	In Desktop OLAP, a user downloads a part of the data from the database locally, or on their and analyze it. DOLAP is relatively cheaper to deploy as it offers very few functionalities compares to othe systems.
Web OLAP (WOLAP)	Web OLAP which is OLAP system accessible via the web browser. WOLAP is a thr architecture. It consists of three components: client, middleware, and a database server.
Mobile OLAP:	Mobile OLAP helps users to access and analyze OLAP data using their mobile devices
Spatial OLAP :	SOLAP is created to facilitate management of both spatial and non-spatial data in a Ge Information system (GIS)

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ROLAP

ROLAP works with data that exist in a relational database. Facts and dimension tables are stored as relational tables. It also allows multidimensional analysis of data and is the fastest growing OLAP.

Advantages of ROLAP model:

- **High data efficiency.** It offers high data efficiency because query performance and access language are optimized particularly for the multidimensional data analysis.
- **Scalability.** This type of OLAP system offers scalability for managing large volumes of data, and even when the data is steadily increasing.

Drawbacks of ROLAP model:

- **Demand for higher resources:** ROLAP needs high utilization of manpower, software, and hardware resources.
- **Aggregately data limitations.** ROLAP tools use SQL for all calculation of aggregate data. However, there are no set limits to the for handling computations.
- **Slow query performance.** Query performance in this model is slow when compared with MOLAP

MOLAP

MOLAP uses array-based multidimensional storage engines to display multidimensional views of data. Basically, they use an OLAP cube.

Learn more about MOLAP here

Hybrid OLAP

Hybrid OLAP is a mixture of both ROLAP and MOLAP. It offers fast computation of MOLAP and higher scalability of ROLAP. HOLAP uses two databases.

1. Aggregated or computed data is stored in a multidimensional OLAP cube
2. Detailed information is stored in a relational database.

Benefits of Hybrid OLAP:

- This kind of OLAP helps to economize the disk space, and it also remains compact which helps to avoid issues related to access speed and convenience.

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- Hybrid HOLAP's uses cube technology which allows faster performance for all types of data.
- ROLAP are instantly updated and HOLAP users have access to this real-time instantly updated data. MOLAP brings cleaning and conversion of data thereby improving data relevance. This brings best of both worlds.

Drawbacks of Hybrid OLAP:

- Greater complexity level**:** The major drawback in HOLAP systems is that it supports both ROLAP and MOLAP tools and applications. Thus, it is very complicated.
- Potential overlaps**:** There are higher chances of overlapping especially into their functionalities.

Advantages of OLAP

- OLAP is a platform for all type of business includes planning, budgeting, reporting, and analysis.
- Information and calculations are consistent in an OLAP cube. This is a crucial benefit.
- Quickly create and analyze "What if" scenarios
- Easily search OLAP database for broad or specific terms.
- OLAP provides the building blocks for business modeling tools, Data mining tools, performance reporting tools.
- Allows users to do slice and dice cube data all by various dimensions, measures, and filters.
- It is good for analyzing time series.
- Finding some clusters and outliers is easy with OLAP.
- It is a powerful visualization online analytical process system which provides faster response times

Disadvantages of OLAP

- OLAP requires organizing data into a star or snowflake schema. These schemas are complicated to implement and administer
- You cannot have large number of dimensions in a single OLAP cube
- Transactional data cannot be accessed with OLAP system.
- Any modification in an OLAP cube needs a full update of the cube. This is a time-consuming process

CASE STUDIES:

This case study is focusing on building and implementing data warehouse concerning companies 'profitability. As data growth and demands grew too great for the company's legacy system, it was falling short in delivering the information which the managers and administrators required on a daily basis. Companies still using out-dated software like central reservation system (CRS) that took almost 18 hours a day to report however, such delay has prevented managers from adjusting rates appropriately, which could result in lost revenue. With a new implementation

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of current data warehouse for example Oracle Exada can complete extract, transform, and load (ETL) operations for operational reports in 4 to 6 hours, as opposed to 18 to 24 hours previously, a six-fold improvement.

MULTIPLE CHOICE QUESTIONS (MCQS): (1 MARKS EACH):

1. What is a Data Warehouse?

A. A database designed to handle transactions.

B. A database designed to support analytical reporting.

C. A database designed to support operational reporting.

D. A database designed to support transactional processing.

Answer: B. A database designed to support analytical reporting.

2. What is ETL?

A. Extract, Transform, Load.

B. Extract, Translate, Load.

C. Extract, Transfer, Load.

D. Extract, Transcribe, Load.

Answer: A. Extract, Transform, Load.

3. What is the difference between OLTP and OLAP?

A. OLTP is optimized for transaction processing, while OLAP is optimized for analytical processing.

B. OLTP is optimized for analytical processing, while OLAP is optimized for transaction processing.

C. OLTP and OLAP are the same thing.

D. OLTP and OLAP are both optimized for transaction processing.

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Answer: A. OLTP is optimized for transaction processing, while OLAP is optimized for analytical processing.

4. What is a fact table?

A. A table that stores details about customers.

B. A table that stores details about products.

C. A table that stores transactional data.

D. A table that stores metadata.

Answer: C. A table that stores transactional data.

5. What is a dimension table?

A. A table that stores transactional data.

B. A table that stores metadata.

C. A table that stores details about customers.

D. A table that stores details about products.

Answer: D. A table that stores details about products.

6. What is a star schema?

A. A schema where all tables are connected to each other.

B. A schema where there is no connection between tables.

C. A schema where one fact table is connected to multiple dimension tables.

D. A schema where multiple fact tables are connected to one dimension table.

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Answer: C. A schema where one fact table is connected to multiple dimension tables.

7. What is an OLAP cube?

A. A database that is optimized for handling transactions.

B. A database that is optimized for analytical reporting.

C. A database that is optimized for operational reporting.

D. A database that is optimized for both analytical and operational reporting.

Answer: B. A database that is optimized for analytical reporting.

8. What is a snowflake schema?

A. A database schema where all tables are connected to a single central table.

B. A database schema where all tables are connected to a single fact table.

C. A database schema where the dimension tables are normalized into multiple related tables.

D. A database schema where multiple fact tables are connected to one dimension table.

Answer: C. A database schema where the dimension tables are normalized into multiple related tables.

9. What is OLAP?

A. A type of database designed for online transaction processing.

B. A type of database designed for online analytical processing.

C. A type of database designed for data warehousing.

D. A type of database designed for data mining.

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Answer: B. A type of database designed for online analytical processing.

10. What is a data cube?

A. A database design that uses a central fact table and related dimension tables to represent data.

B. A type of database that is optimized for data mining and predictive analytics.

C. A type of database that is designed for online analytical processing.

D. A multi-dimensional representation of data used in OLAP applications.

Answer: D. A multi-dimensional representation of data used in OLAP applications.

SHORT ANSWER QUESTIONS (3 MARKS EACH):

1. Define data warehouse.
2. What is the need of data warehouses?
3. Define OLAP.
4. Define multidimensional data model.
5. What is a data cube?
6. Define dimensions.
7. What are facts?
8. Define OLTP.
9. Define OLAP.
10. Define dimension table.
11. Define fact table.
12. List out the various OLAP operations.
13. Give the names of warehouse schemas.
14. Define star schema.
15. Define snowflake schema.
16. Draw a neat diagram of data warehouse architecture.

LONG ANSWER QUESTIONS (5 MARKS EACH):

1. Explain multidimensional data model with a neat diagram.
2. List out the OLAP operations and explain the same with an example.

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3. Describe about dimension modeling in detail.
4. Explain the various schemas of a data warehouse.
5. Define data warehouse. Draw the architecture of data warehouse and explain the three tiers in detail.
6. Explain in detail about the implementation of a data warehousing.
7. Define metadata and explain the types of metadata.
8. Discuss the development lifecycle of a data warehouse.
9. Explain the processes taking place in the backend of a data warehouse.

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